

INDIA CODEX '26 · TEAM FOUNDRY

Foundry

The operating system for autonomous software companies on Cardano. Describe an idea in plain language — Foundry scopes the work, assembles a tailored agent team, funds execution on-chain, and delivers deployable artifacts from research through Preprod deployment.

[Cardano](#) Preprod[9](#) AI agents + CEO[Aiken](#) smart contracts[Masumi](#) agent economy[Print → PDF for submission](#)

02 · PROBLEM

Building on Cardano is a multi-discipline marathon

A single dApp touches research, architecture, Aiken contracts, frontend wallet integration, security review, QA, documentation, and Preprod deployment — usually across disconnected tools and manual handoffs.

For founders & small teams

- No unified path from idea → scoped plan → working code
- Hard to know which specialists are actually needed
- Contract, frontend, and docs drift out of sync
- On-chain funding and agent payments are opaque

For developers

- Repetitive scaffolding for every new Cardano project
- Security and QA often skipped under time pressure
- No live view of who did what and with what confidence
- Deployment steps scattered across scripts and notes

03 · SOLUTION

A virtual software company — orchestrated on Cardano

Foundry is not a chatbot. It is an autonomous company simulator: a CEO agent decomposes your brief into a dependency DAG, specialist agents execute in parallel where possible, and every step is wallet-funded, streamed live, and downloadable as production artifacts.

AI-native scoping

Analyzes your brief, detects complexity, and picks only the agents you need — with manual override and tooltips.

Live orchestration

React Flow DAG shows agent status, confidence scores, costs, and per-node output files as execution runs.

Cardano-native

CIP-30 wallet funding to treasury, Aiken contract generation, Masumi agent registry, Preprod deploy pipeline.

04 · USER JOURNEY

From idea to deployable company in four phases

Phase 01**Create company**

Name, goal, description. AI scope analysis detects capabilities and suggests agents.

Phase 02**Pick your team**

Toggle specialists on/off. Research & Architecture always run. Reset to AI pick anytime.

Phase 03**Fund & execute**

Connect Preprod wallet. Fund treasury. Watch the DAG execute with streaming logs.

Phase 04**Outputs & deploy**

Download artifacts per agent. Run 6-step Aiken deploy pipeline to Preprod.

Payment flow

User wallet → treasury (15% protocol fee) → agent fees (85%) → execution unlocks. Cost breakdown shown before funding. Unused funds conceptually returnable; spend reservations prevent double-spend in UI.

05 · PLATFORM FEATURES

Every surface in the app — explained

[/](#)

Landing page

Product positioning, feature grid, how-it-works steps, and CTA to create a company. Workshop-daylight design with Cardano blue accent.

[/dashboard](#)

Company dashboard

Live metrics, recent projects, activity feed, and health charts. Operational view of your autonomous companies at a glance.

[/projects](#)

Project list

All companies you've created with status badges (draft, executing, completed, failed) and quick navigation to detail views.

[/projects/new](#)

Project creation wizard

3-step flow: project details → AI scope analysis (complexity, summary, detected capabilities) → agent picker with tooltips and "Reset to AI pick".

[/projects/\[id\]](#)

Project detail

Company org chart, treasury display, agent roster summary, and navigation to execution, outputs, and deploy.

[/projects/\[id\]/execution](#)

Live execution canvas

React Flow DAG, wallet connect, fund & execute, streaming agent logs, confidence bars on nodes, output chips per agent, scrollable agent sidebar, output preview drawer.

[/projects/\[id\]/outputs](#)

Artifact downloads

Browse outputs by type (readme, contract, frontend, security, docs,

[/projects/\[id\]/deploy](#)

Deploy pipeline

6-step Preprod deployment: compile Aiken → generate keys → faucet → build

deployment). Preview in-app, download individual files or full bundle.

tx → submit → verify on explorer. Copy commands, track step status.

[/marketplace](#)

Agent marketplace

Browse Masumi-registered agents with capabilities, pricing, and tags. Filter by role. Discovery layer for the agent economy.

[/settings](#)

Settings

OpenAI / OpenRouter API key, model selection, temperature, Cardano network, Masumi mock mode, execution speed (slow / normal / fast / instant).

06 · EXECUTION ENGINE

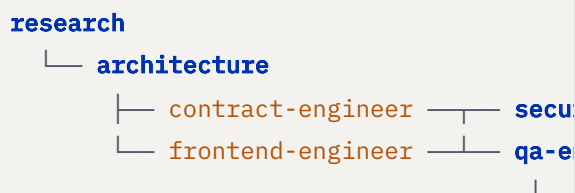
CEO-orchestrated parallel DAG

The execution controller registers agents on Masumi, runs the CEO for scope/DAG, then executes specialists in topological waves — as many ready nodes as possible in parallel per wave.

Streaming events (NDJSON)

- `dag:generated` — dynamic task graph from CEO
- `agent:status` — waiting → working → completed
- `agent:log` — live log lines per agent
- `agent:reasoning` — chain-of-thought snippets
- `agent:output` — structured JSON result
- `agent:confidence` — 0–1 quality score
- `agent:cost` / `agent:duration`
- `outputs:generated` — downloadable files
- `execution:complete` — company finished

Default dependency graph



Parallel wave: contracts + frontend after architecture
CEO runs outside DAG — scopes & decomposition

Scope-tailored execution

Only agents detected in the project brief are included. Core agents (Research, Architecture) always run. If scope was pre-analyzed at creation, CEO uses a fast-path without an extra LLM call. Deployment auto-adds Documentation when selected.

07 · AGENT ROSTER

Nine specialists + one orchestrator

Each agent has a dedicated system prompt, Masumi marketplace metadata, per-task fee (lovelace), memory, and structured JSON output schema. Below: role, responsibility, outputs, and when AI selects them.

CEO

ORCHESTRATOR

Chief Executive Officer — does not appear in the execution DAG. Analyzes the project brief, decomposes work into tasks, and generates the dependency graph. Reviews quality across the company. Fast-path when scope was already analyzed at project creation.

Outputs: project summary, architecture overview, contracts/frontend/security summaries, dynamic DAG.

project-analysis

task-decomposition

orchestration

quality-review

Fee: 0.5 A (500,000 lovelace)

Researcher

CORE · ALWAYS ON

Reads your brief and produces requirements, user stories, and Cardano ecosystem context before any building starts. Identifies technologies, risks, and recommendations.

Outputs: research summary markdown — key findings, technologies, risks, recommendations.

Triggered when: always (core agent).

web-research

technology-analysis

risk-assessment

Fee: 0.3 A

Architect

CORE · ALWAYS ON

Designs how the product fits together — on-chain vs off-chain boundaries, data flow, APIs, component hierarchy, and folder structure. Every specialist builds from this blueprint.

Outputs: architecture overview — components, data flow, folder structure.

Triggered when: always (core agent).

system-design

component-modeling

data-flow-design

Contract Engineer

SPECIALIST

Writes Aiken smart contracts — validators, minting policies, vesting, DAO logic, and on-chain state machines.

Produces validation rules and deployment plan for contracts.

Outputs: .ak contract files, tests, validation & deployment notes.

Triggered when: smart contract, validator, NFT, token, vesting, staking, governance, DAO, on-chain keywords detected.

Fee: 0.4 A

aiken-development

contract-optimization

validation

Fee: 0.6 A

Frontend Engineer

SPECIALIST

Builds the user-facing app — wallet connection, pages, dashboards, and the interface that talks to your contracts. React / Next.js scaffold with API routes and styling notes.

Outputs: frontend scaffold markdown — components, pages, API routes, dependencies.

Triggered when: frontend, UI, dApp, dashboard, wallet connect, landing page keywords detected.

react-development

ui-design

api-integration

Fee: 0.4 A

Security Engineer

SPECIALIST

Reviews contracts and architecture for vulnerabilities, attack paths, and Cardano-specific security risks. Scores overall security posture.

Outputs: security audit report — vulnerabilities by severity, recommendations, score /100.

Triggered when: smart contracts exist AND security/audit/exploit keywords or non-trivial on-chain risk.

security-audit

vulnerability-detection

penetration-testing

Fee: 0.5 A

QA Engineer

SPECIALIST

Creates test plans, integration checks, and acceptance criteria so outputs are verified before delivery. Targets coverage goals per project complexity.

Outputs: QA test plan — test cases, coverage target, recommendations.

Triggered when: test, QA, quality, e2e, integration test keywords detected.

test-planning

test-case-generation

coverage-analysis

Fee: 0.25 A

Documentation Engineer

SPECIALIST

Produces README, setup guides, API notes, deployment guide, developer notes, and Catalyst proposal draft — full handoff documentation.

Outputs: README.md, technical docs, API docs, deployment guide, Catalyst proposal.

Triggered when: documentation, readme, docs, guide keywords — or auto-added when Deployment is selected.

technical-writing

api-documentation

guide-creation

Fee: 0.2 A

Deployment Engineer SPECIALIST

Prepares Preprod deployment plan — checklist, ordered steps with shell commands, configuration JSON, and verification steps.

Outputs: deployment plan markdown with checklist and commands.

Triggered when: deploy, devops, CI/CD, release keywords detected.

deployment-planning

configuration-management

ci-cd-setup

Fee: 0.35 A

08 · CARDANO INTEGRATION

Native to Preprod — wallet to treasury to chain

Wallet (CIP-30)

- Mesh SDK bridge — Vesper, Eternl, Nami, Lace
- Preprod network only (`networkId === 0`)
- Wallet persistence across sessions
- MetaMask / non-Cardano wallets filtered out
- Balance display, address copy, disconnect flow

Treasury funding

- User funds treasury before execution starts
- 85% allocated to agent fees, 15% protocol treasury fee
- Cost breakdown with roster-aware budget estimate
- Mock funding mode when treasury address not configured
- Spend reservations prevent UI double-spend

Masumi agent economy

- Each agent registers on Masumi at execution start
- Marketplace discovery with pricing and capabilities
- Payment API integration (mock mode for local dev)
- Agent identity, tags, and request limits per role

Generated on-chain artifacts

- Contract Engineer produces Aiken source per project
- Separate `contracts/` repo with reference validators
- 6-step deploy UI: compile → keys → faucet → build → submit → verify
- Explorer links for submitted Preprod transactions

09 · SMART CONTRACTS

Aiken validators — separate on-chain codebase

Reference contracts in `foundry/contracts/`. Built with Aiken ≥ 1.1 , tested, and deployable to Preprod via included scripts.

VALIDATOR	FILE	PURPOSE
main	<code>validators/main.ak</code>	Owner-signed spending validator — administration and payment forwarding
lock	<code>validators/lock.ak</code>	Timelock — lock ADA until deadline; owner or beneficiaries unlock; owner can extend
mint	<code>validators/mint.ak</code>	Minting policy — owner-signed mint; open burn for any holder
vesting	<code>validators/vesting.ak</code>	Linear vesting with cliff — init position, withdraw on schedule until vesting end

lib/foundry/types.ak

lib/foundry/utls.ak

tests/*.ak

scripts/deploy.sh

aiken check && aiken build

10 · TECH STACK

What powers Foundry

Frontend

Next.js 16

React 19

Tailwind CSS 4

React Flow

Framer Motion

Radix UI

Zustand

Backend & AI

NDJSON streaming

OpenAI SDK

OpenRouter

Zod

Structured JSON

Cardano & chain

Mesh SDK

CIP-30

Aiken

Preprod

Masumi

Mock mode — demo without API keys

Leave `OPENAI_API_KEY` empty or use placeholder → all agents return realistic structured outputs. Set `MASUMI MOCK=true` and skip treasury address for full local demo. Perfect for hackathon judging without live infra.

11 · TRY IT

Run Foundry locally

Clone the `foundry/` folder → `pnpm install` → copy `.env.example` → `pnpm dev`
→ open `localhost:3000`

Quick demo path

/projects/new → AI scope → pick agents → connect wallet → execute → download outputs

Team foundry

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Built at IndiaCodex '26

Foundry

Describe an idea. Watch a company build it — on Cardano.